



IBM Developer
SKILLS NETWORK

Understanding Rocket Science with Data Analysis

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Executive Summary

- Summary of methodologies

Scrapped data from SpaceX API, Wikipedia. EDA with SQL and visualization using Python through Jupyter notebooks. Modelling with Scikit-learn, comparing the methods of Regression, KNN, Decision Tree and SVM, using cross validation and grid search to find best hyperparameters.

- Summary of all results

There's a relation between payload mass and launch site with the landing success rate. Best model to predict with high accuracy and precision is the Decision Tree Classification Algorithm, with parameters ..., leading to accuracy

Introduction

- Space X advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage.
- If we can determine if the first stage will land, we can determine the cost of a launch. This information can be used if an alternate company wants to bid against space X for a rocket launch. Main goal: create a machine learning pipeline to predict if the first stage will land given the data from SpaceX previous launches.



Section 1 Methodology

Methodology

Executive Summary

- Data collection methodology:

Requested data through SpaceX API, as web scrapping from Wikipedia, using python packages: requests, BeautifulSoup, pandas, numpy.

- Perform data wrangling

Using pandas mostly, we formatted the data, such as fixing missing of PayloadMass with the mean value and converted the Booster Landing outcomes to a categorical class of 1 = 'success' and 0 = 'fail'.

- Perform exploratory data analysis (EDA) using visualization and SQL

- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models

Using the Scikit-learn python library, we first split the data into training and testing sets, then we train each classification algorithm model: [Logistic Regression](#), [Support Vector Machine](#), [Decision Tree](#), and [K-Nearest Neighbors](#).

The hyperparameters of each model are chosen through a Grid Search and cross-validation. Then the best model accuracy and score on the test data are measured, as the confusion matrix, to find the method that performs the best.

Data Collection

'dataset_part_1.csv' was collected through the SpaceX API <https://api.spacexdata.com/v4/>, making HTTP requests.

'spacex_web_scraped.csv' was collected through web scrapping the Wikipedia page https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches, using HTTP requests.



Data Collection – SpaceX API

Data collection with SpaceX REST calls

- Request access to the server:

```
response = requests.get(spacex_url)
```

- Decode the response into json:

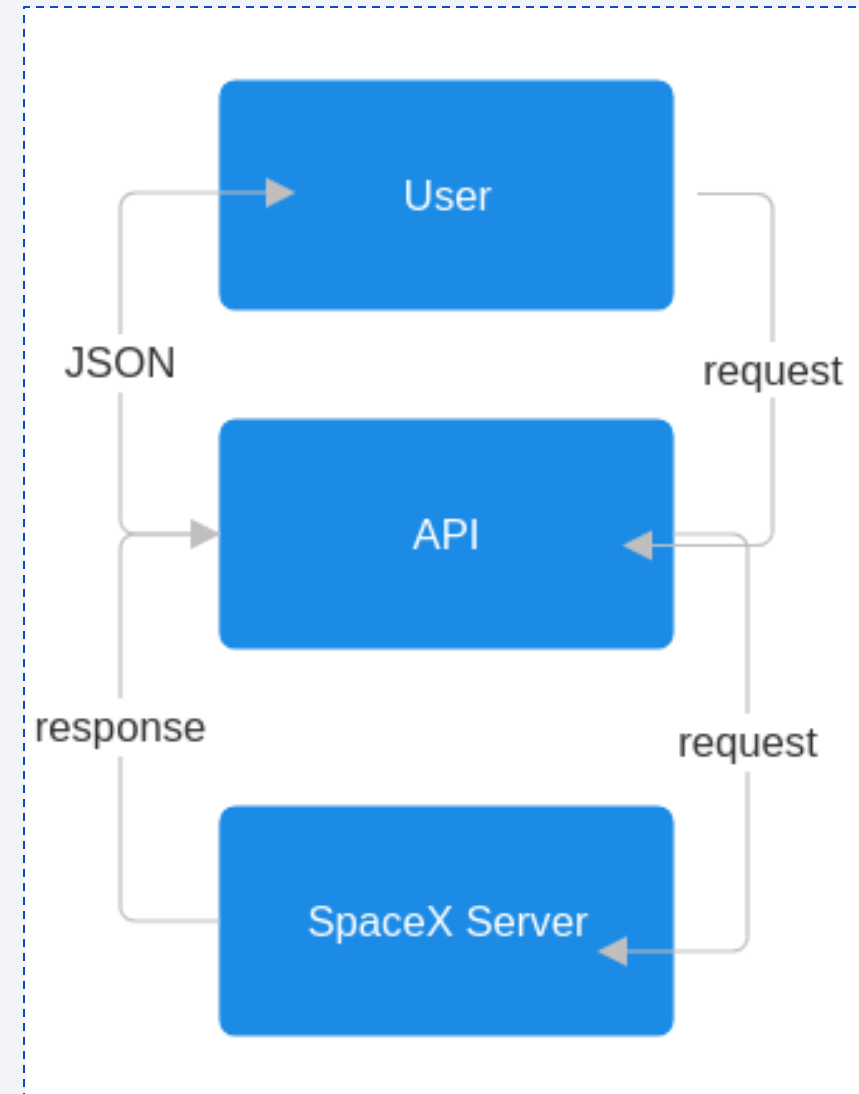
```
decoded = response.json()
```

- Transform json into dataframe:

```
data = pd.json_normalize(decoded)
```

GitHub URL of the completed SpaceX API calls notebook

https://github.com/luksgabo/Capstone_project/blob/main/data_collection.ipynb



Data Collection - Scraping

- Request access to the server

```
response = requests.get(wiki_url, headers=headers)
```

- Create a BeautifulSoup object

```
soup = BeautifulSoup(response.text,  
features="html.parser", )
```

- Find tables

```
html_tables = soup.find_all('table')
```

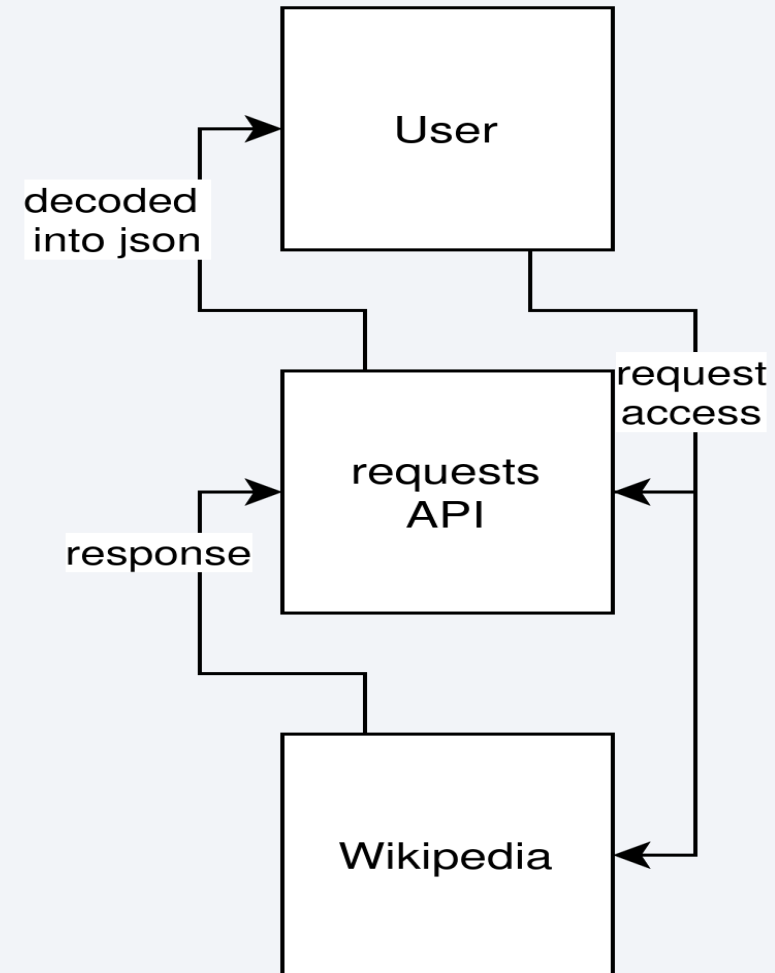
- Decoded to dictionary then to a dataframe.

- Saved to csv file

```
df.to_csv('data/spacex_web_scraped.csv', index=False)
```

- GitHub URL of the completed web scraping notebook

https://github.com/luksgabo/Capstone_project/blob/main/webscrapping.ipynb



Data Wrangling

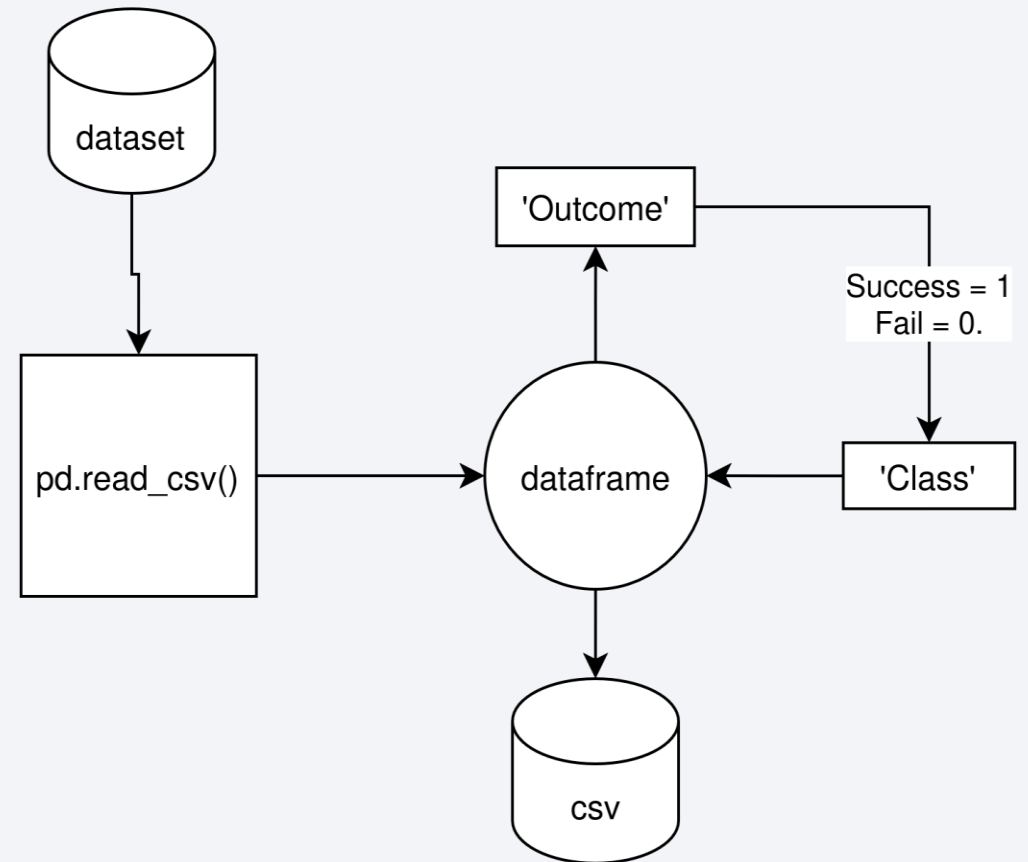
Loaded dataset from previous lab.

Created a column 'Class' with categorical data on the landing success:

```
bad_outcomes = [out for out in  
    set(df.Outcome.values) if 'True' not in out]
```

```
landing_class = [int(out not in  
    bad_outcomes) for out in  
    df.Outcome.values ]
```

- GitHub URL of the data wrangling notebook
https://github.com/luksgabo/Capstone_project/blob/main/data_wrangling.ipynb



EDA with Data Visualization

We plotted:

- scatter plots to visualize the relationships between payload mass, flight number, launch site, and orbit type.
- bars chart to visualize the success rate of each orbit type launch.
- line plot to show the yearly trend of the landings success rate.
- GitHub URL of the completed EDA with data visualization notebook https://github.com/luksgabo/Capstone_project/blob/main/exploratory_data_analysis2.ipynb

EDA with SQL

- Connected to database `con = sqlite3.connect("my_data1.db")`
- Created table `create table SPACEXTABLE as select * from SPACEXTBL where Date is not null`
- Queries to explore the data, such as "Display the names of the unique launch sites in the space mission", "List the total number of successful and failure mission outcomes", etc..

```
%%sql
SELECT DISTINCT(Launch_Site) FROM SPACEXTABLE;
```

```
%%sql
SELECT COUNT(Mission_Outcome) AS 'Total missions'
FROM SPACEXTABLE;
```

- GitHub URL of completed EDA with SQL notebook https://github.com/luksgabo/Capstone_project/blob/main/exploratory_data_analysis.ipynb

Build an Interactive Map with Folium

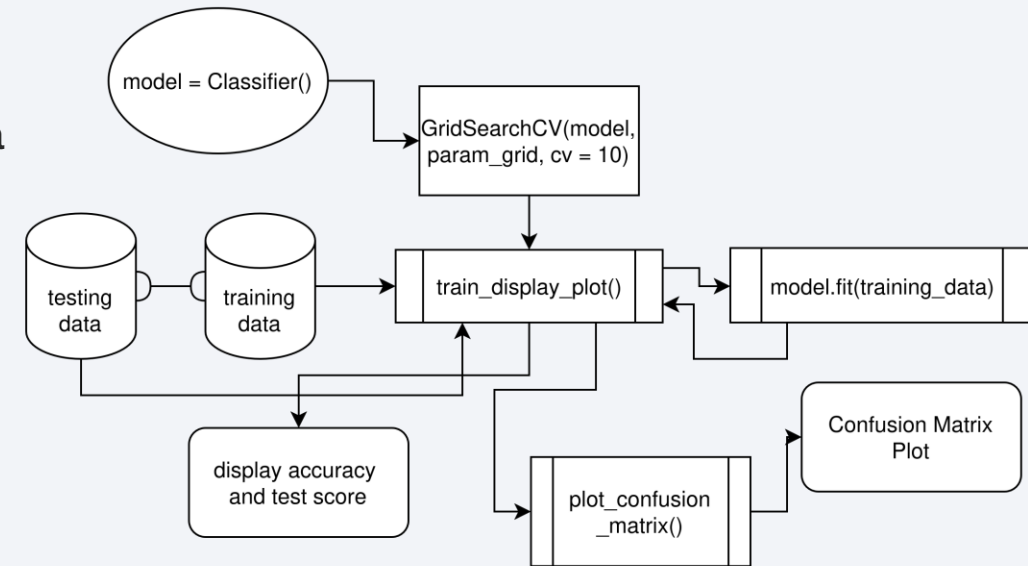
- Circles and markers: NASA location, Launch sites, clusters of launches with information about landing success. To understand the relation between location and the rocket launches.
- Lines: distances of launch sites to nearby roads, cities, shores, etc..
- GitHub URL of completed interactive map with Folium map https://github.com/luksgabo/Capstone_project/blob/main/launch_sites_visualization.ipynb

Build a Dashboard with Plotly Dash

- Plots/graphs and interactions added to the dashboard
 - dropdown list to enable Launch Site selection
 - pie chart to show the total successful launches count for Launch Site selected
 - slider to select payload range
 - scatter chart to show the correlation between payload and launch success
- GitHub URL of completed Plotly Dash lab https://github.com/luksgabo/Capstone_project/blob/main/spacex-dash-app.ipynb

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
 - Built: Train model using part of the data
 - Evaluated: Cross-validation, by splitting the training data into a validation set and checking accuracy
 - Improved: Grid Search of hyperparameters combinations to get maximum accuracy
 - Final score evaluating the model on the test data.
- GitHub URL of completed predictive analysis lab https://github.com/luksgabo/Capstone_project/blob/main/prediction_model.ipynb



Results

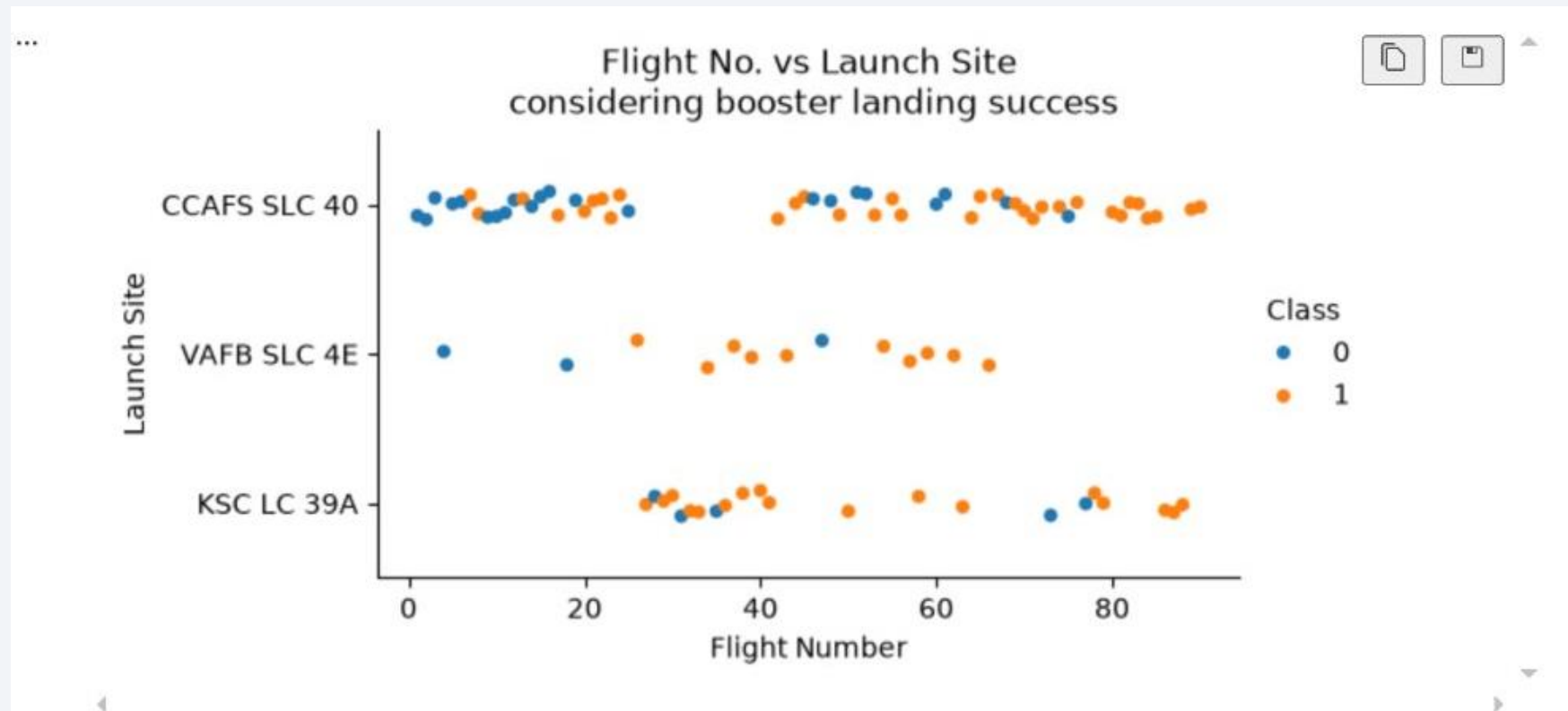
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

A magnifying glass is positioned over a bar chart. The chart features blue and green bars for three quarters labeled Q2, Q3, and Q4. A text box in the upper left corner of the magnified area contains the text 'Section 2' and 'Insights from EDA'. A '1,000' scale marker is visible on the right side of the chart.

Section 2

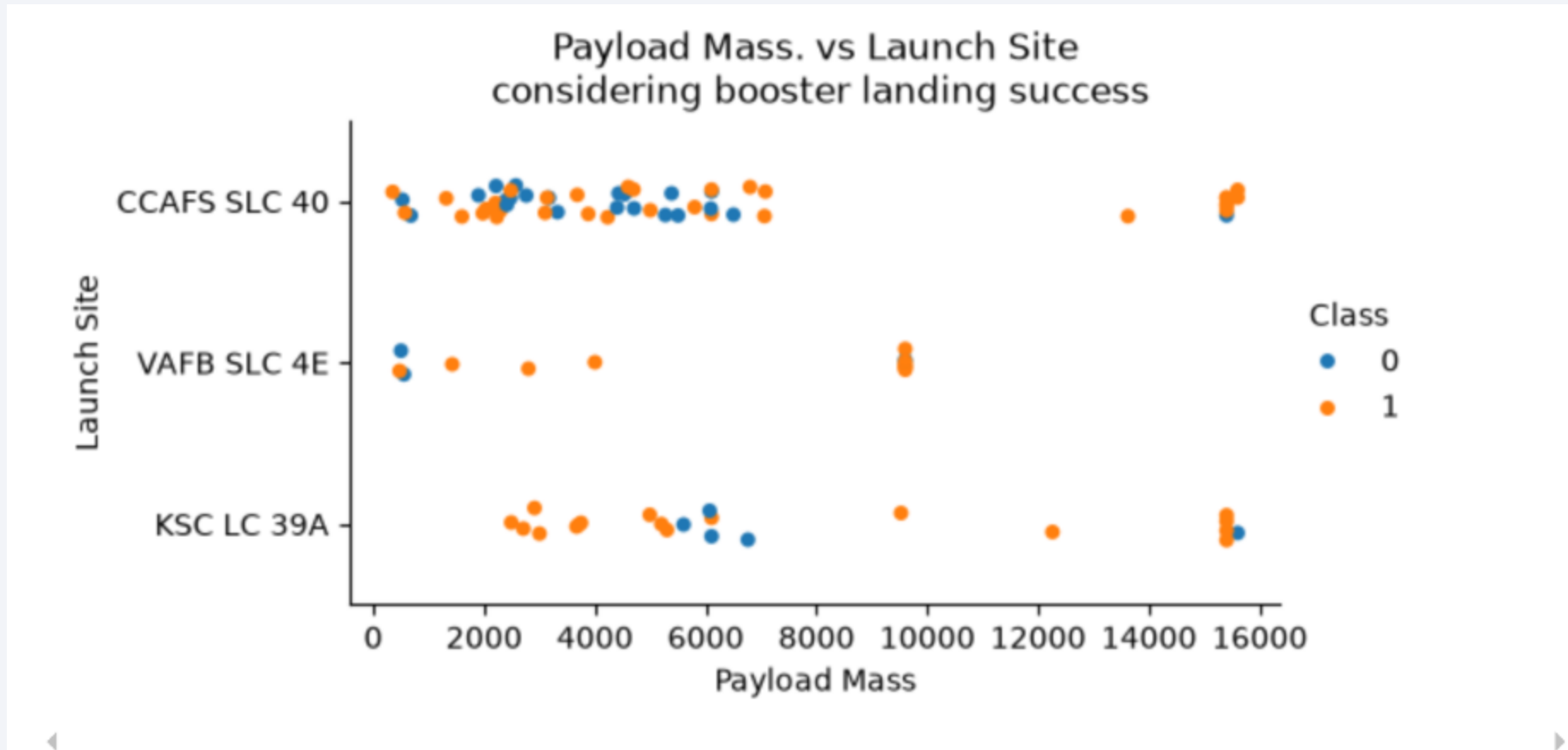
Insights from EDA

Flight Number vs. Launch Site



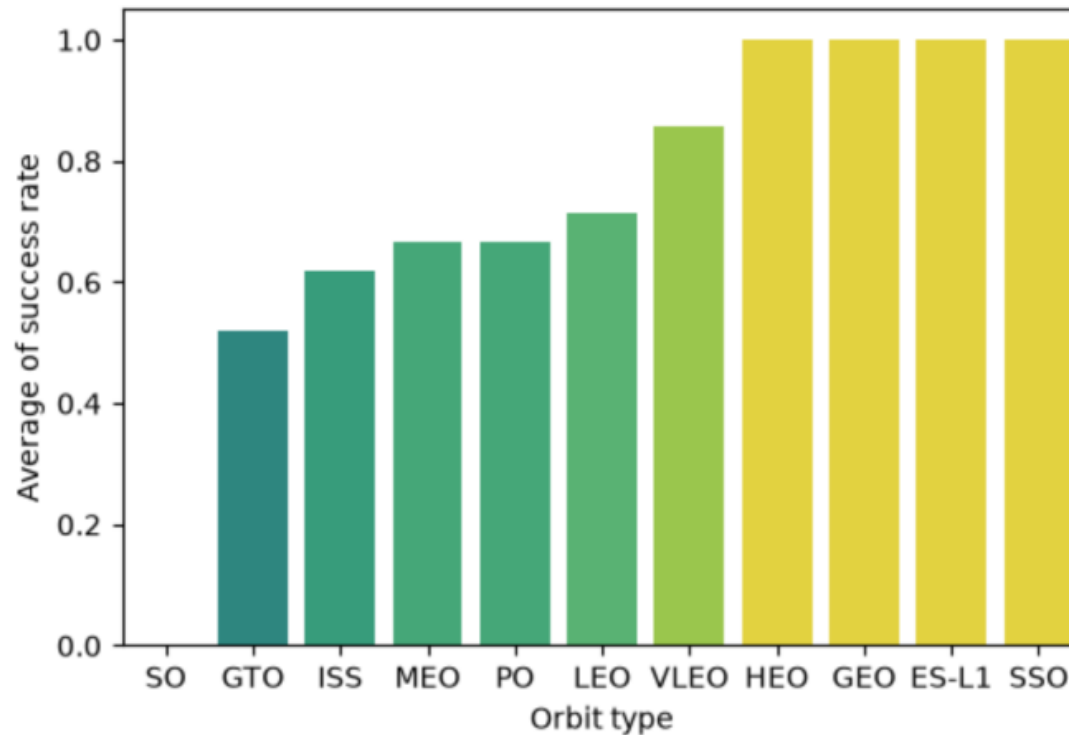
Grouping the data into Launch Sites we can see that for higher flight number the success rate (class = 1) increases. Probably because the company learned with each launching and improved the landing success after. We can see also that VAFB and KSC sites have a higher ratio of successful landings than CCAFS.

Payload vs. Launch Site



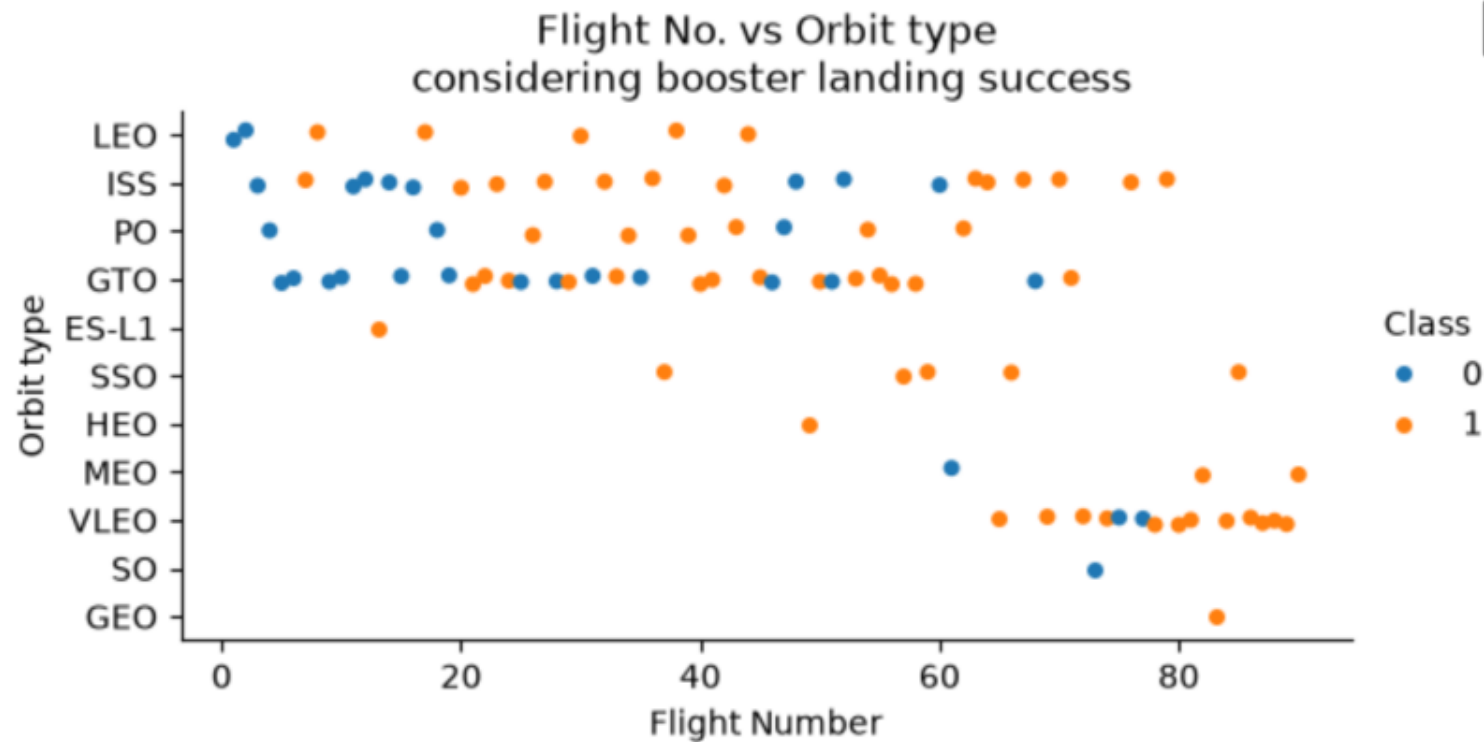
From the plot we can see that higher payload masses are correlated to higher success rates for landing, specially for masses over 8000kg.

Success Rate vs. Orbit Type



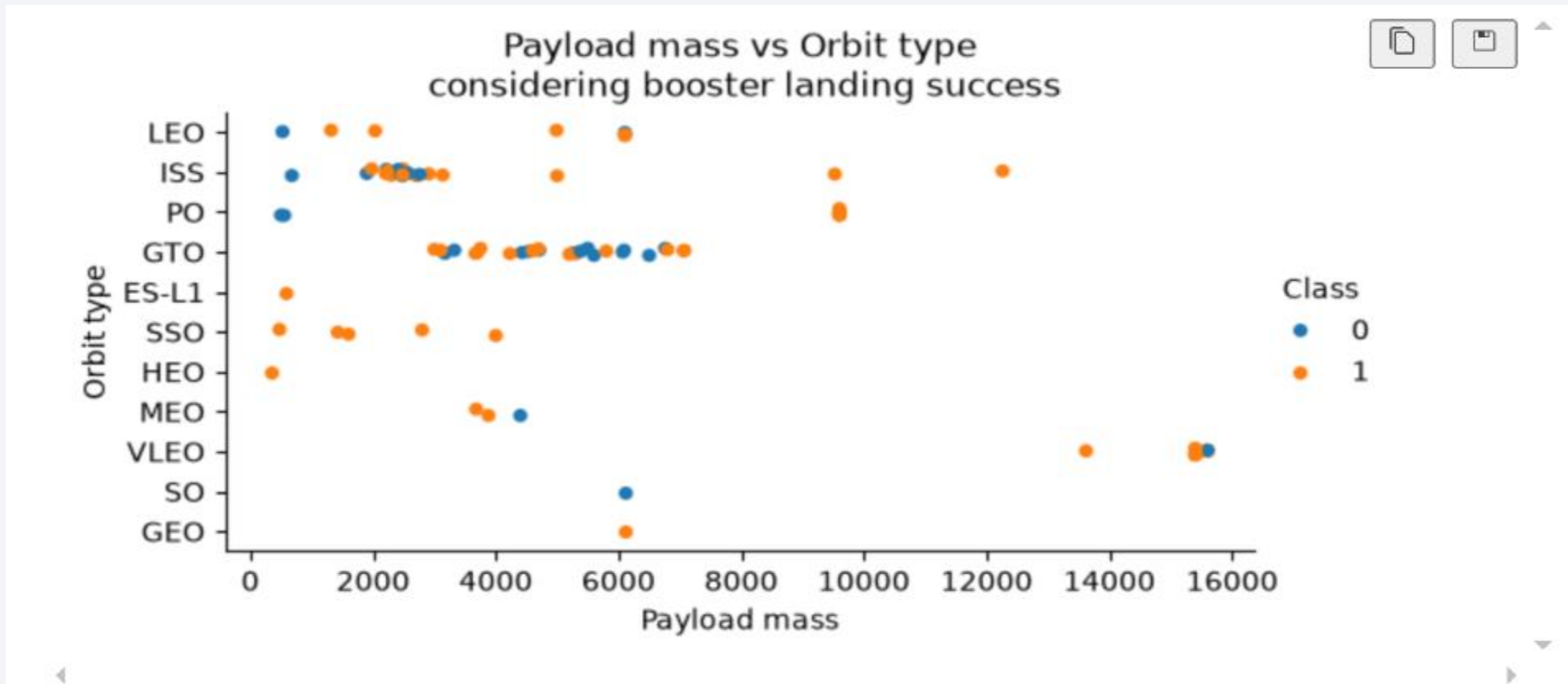
For 4 Orbit types we have a average success rate of 100% (HEO, GEO, ES-L1 and SSO), while for other types the averages are between 50% and 85%. For SO orbits we only have one record, with a failed landing, which explains the 0% average success rate. We would need to check the standard deviation for each orbit type to understand the data better.

Flight Number vs. Orbit Type



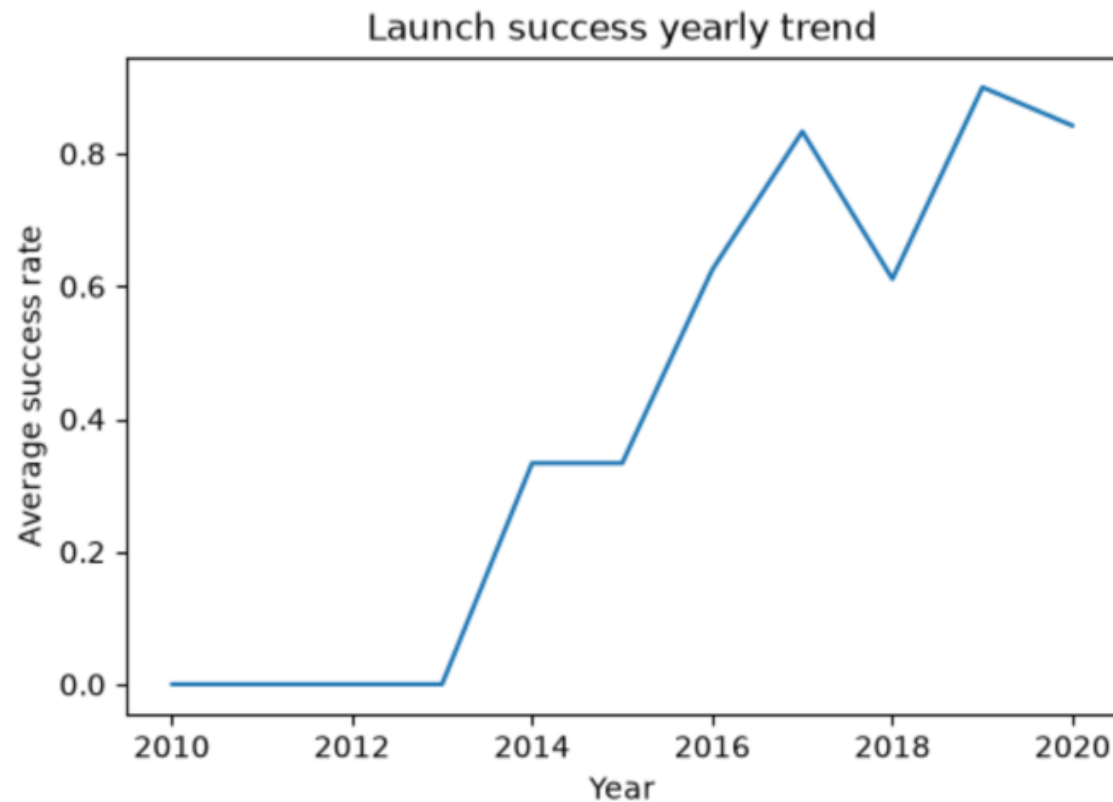
A pattern we can spot in the plot is that landing success rate increases with the flight number for each orbit type. Probably because the company learns with previous failures. Some orbit types also show higher success ratios to others, such SSO and VLEO.

Payload vs. Orbit Type



We can see that for each orbit type, the success rate increases with the payload mass, as failed landings are mostly in low masses ranges. But for GTO, MEO and VLEO orbits that's not true, and the first type of orbit has a well distributed failures and successes.

Launch Success Yearly Trend



The company increased the landings success rate every year from 2013 and over, with some small reductions in 2018 and 2020. But since 2015 it has increased from 50% to over 80%.

All Launch Site Names

- Query:
 - %%sql
SELECT DISTINCT(Launch_Site) FROM SPACEXTABLE;
- Result:

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40
- The DISTINCT function gets the unique values of the column 'Launch_Site' from the table SPACEXTABLE

Launch Site Names Begin with 'CCA'

- Query:
 - %%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
- Result:
see figure!
- The LIKE function gets data that follows the pattern 'CCA%' where % means anything else after. LIMIT shows only a limited amount of data.

* [sqlite:///my_data1.db](#)

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)

Total Payload Mass

- Query:
 - %%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS 'payload mass sum for customer NASA (CRS)
in kg'
FROM SPACEXTABLE
WHERE Customer LIKE 'NASA (CRS)';
- Result:
 - payload mass sum for customer NASA (CRS) in kg: 45596
- Function SUM gets the sum of all data in the column selected, AS changes the columns name. The WHERE condition selects just data from 'Customer' that matches 'NASA (CRS)', the desired customer.

Average Payload Mass by F9 v1.1

- Query:
 - %%sql
SELECT AVG(PAYLOAD_MASS__KG_) AS 'mean payload mass for booster version F9 v1.1 in kg'
FROM SPACEXTABLE
WHERE Booster_Version LIKE 'F9 v1.1';
- Result:
 - mean payload mass for booster version F9 v1.1 in kg: 2928.4
- The AVG function gets the mean of all values in the selected column, with the WHERE condition that selects only Booster Version of 'F9 v1.1'.

First Successful Ground Landing Date

- Query:
 - %%sql
SELECT MIN(DATE)
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE '%Success%ground pad%';
- Result:
 - MIN(DATE): 2015-12-22
- Function MIN gets the minimum value of the selected column 'DATE', using the condition of 'Landing_Outcome' matching the pattern '%Success%ground pad%' which selects only the successful landing in ground land.

Successful Drone Ship Landing with Payload between 4000 and 6000

- Query:

- %%sql

```
SELECT DISTINCT(Booster_Version)
FROM SPACEXTABLE
WHERE LOWER(Landing_Outcome) LIKE '%success%drone ship%'
AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000;
```

- Result:

- Booster_Version: F9 FT B1022; F9 FT B1026; F9 FT B1021.2; F9 FT B1031.2

- Select unique Booster Versions from the table, with conditions on the Landing Outcome to be a successful drone ship land, and that the payload mass values are in between 4000 and 6000.

Total Number of Successful and Failure Mission Outcomes

- Query:
 - %%sql
SELECT COUNT(Mission_Outcome) AS 'Successful missions'
FROM SPACEXTABLE
WHERE LOWER(Mission_Outcome) LIKE '%success%';
 - %%sql
SELECT COUNT(Mission_Outcome) AS Failed missions'
FROM SPACEXTABLE
WHERE LOWER(Mission_Outcome) LIKE '%failure%';
- Result:
 - Successful missions = 100, Failed missions = 1.
- Function COUNT gets the occurrences of data in the columns that follows the condition on the Mission Outcome data.

Boosters Carried Maximum Payload

- Query:

- %%sql

```
SELECT DISTINCT Booster_Version, PAYLOAD_MASS__KG_ AS 'MAX  
PAYLOAD_MASS__KG_'  
FROM SPACEXTABLE  
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM  
SPACEXTABLE);
```

- Result:

- Booster_Version: F9 B5 B1048.4; F9 B5 B1049.4; F9 B5 B1051.3; F9 B5 B1056.4; F9 B5 B1048.5; F9 B5 B1051.4; F9 B5 B1049.5; F9 B5 B1060.2; F9 B5 B1058.3; F9 B5 B1051.6; F9 B5 B1060.3; F9 B5 B1049.7

- With a subquery to get the maximum payload mass, we select unique values of Booster Version such that their payload mass are equal to the max value of 15600 kg.

2015 Launch Records

- Query:

- %%sql
SELECT
CASE SUBSTR(DATE,6,2)
 WHEN '01' THEN 'January' WHEN '02' THEN 'February' WHEN '03' THEN 'March'
 WHEN '04' THEN 'April' WHEN '05' THEN 'May' WHEN '06' THEN 'June'
 WHEN '07' THEN 'July' WHEN '08' THEN 'August' WHEN '09' THEN 'September'
 WHEN '10' THEN 'October' WHEN '11' THEN 'November' WHEN '12' THEN 'December'
END
AS MONTH, LANDING_OUTCOME, BOOSTER_VERSION, LAUNCH_SITE
FROM SPACEXTABLE
WHERE SUBSTR(DATE,0,5)='2015'
AND LOWER(LANDING_OUTCOME) LIKE '%failure%ship%';

- Result: Using case to change the month value to its name.

MONTH	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Query:

- %%sql
SELECT LANDING_OUTCOME, COUNT(*) AS outcome_count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY LANDING_OUTCOME
ORDER BY outcome_count DESC;

- Results:

- See table! No attempts of landing have the highest counts, of 10, followed by success, and failures at drone ships, each with 5 counts, probably because it is easier and safer to land in the sea. The least tried landing outcome is a failed precluded at a drone ship.

Landing_Outcome	outcome_count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

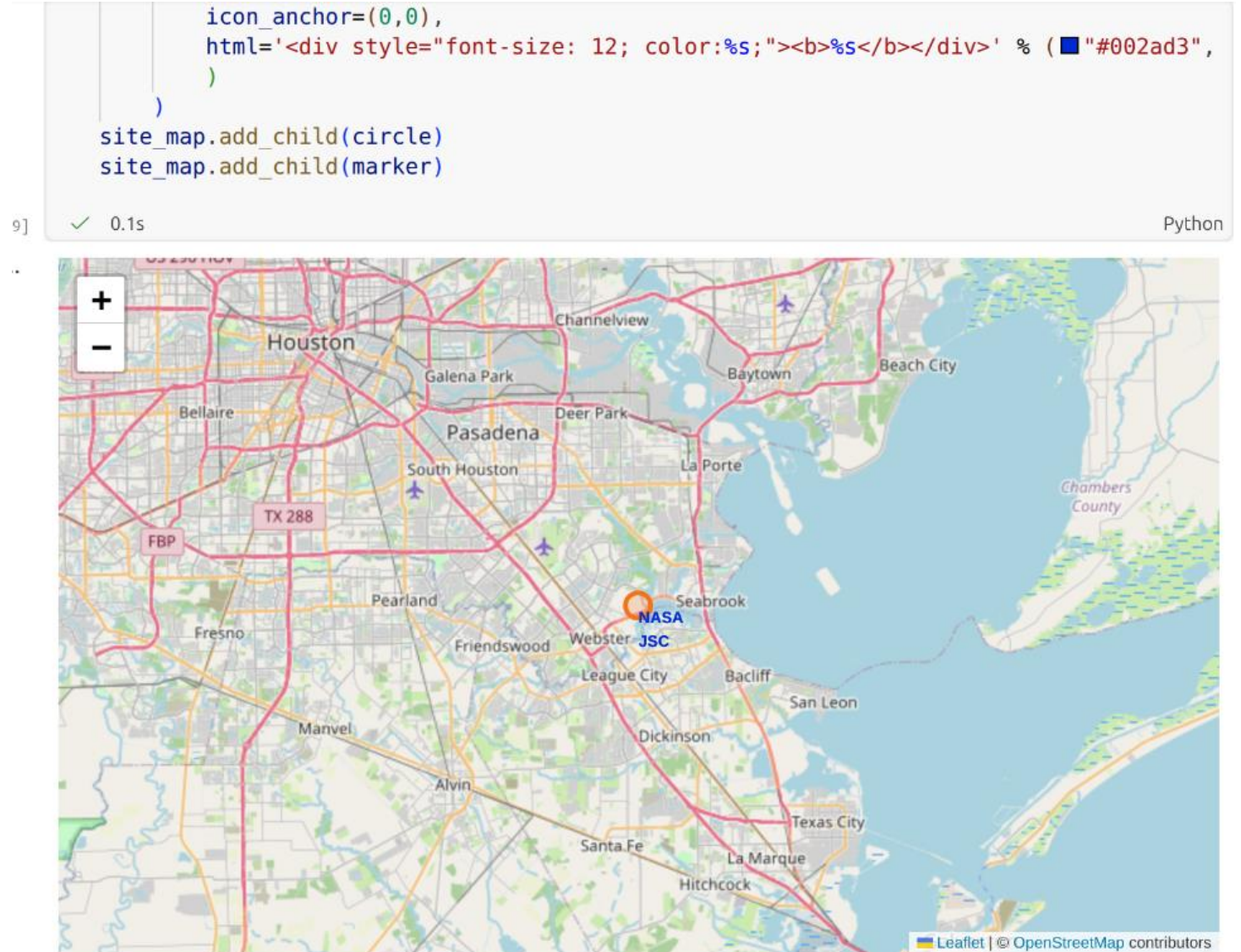
Section 3

Launch Sites proximities



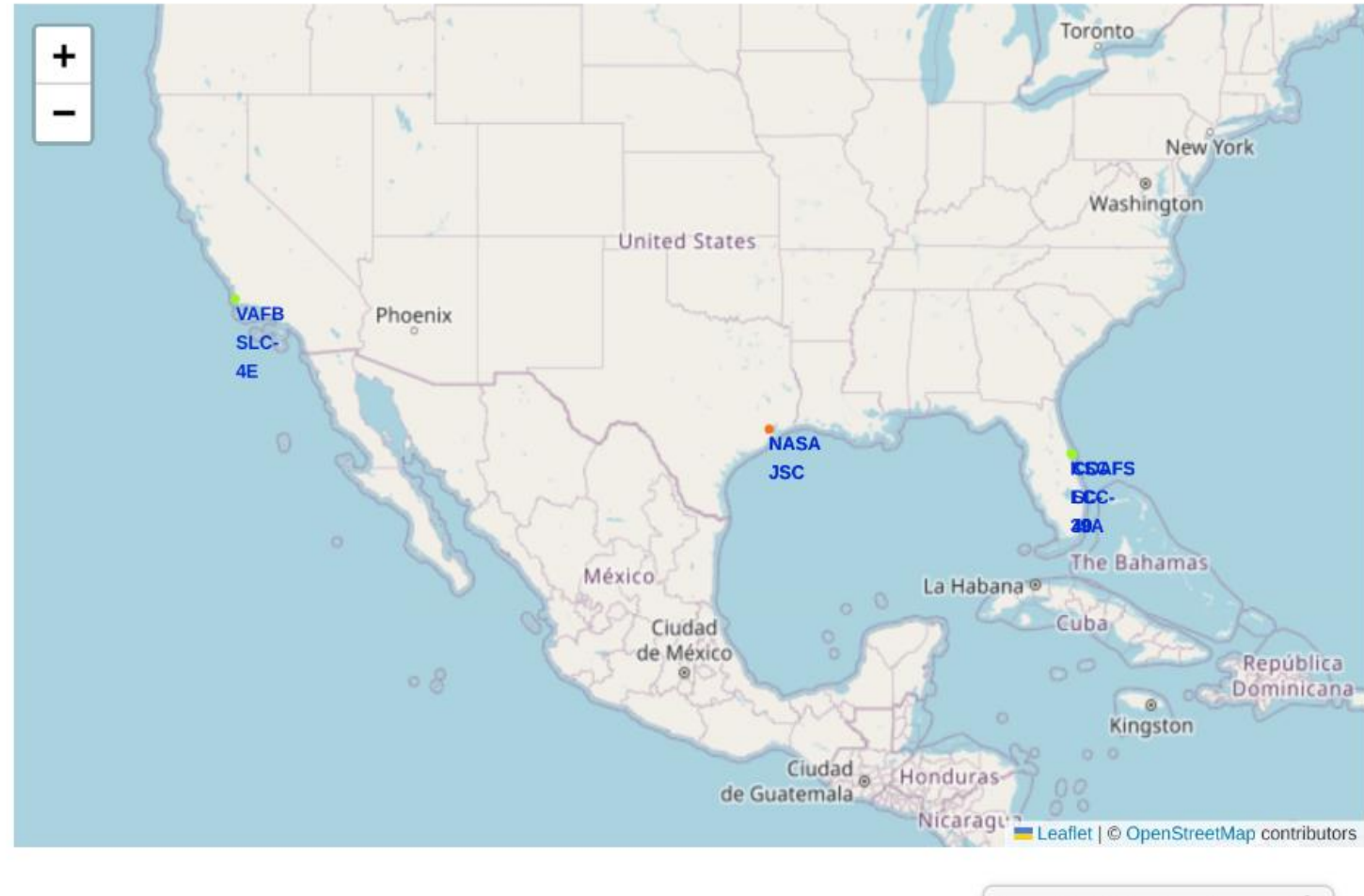
NASA location on Folium map

- Added a circle with the NASA location
- The map shows Houston in the upper left and Nasa in the center near the bay shore.



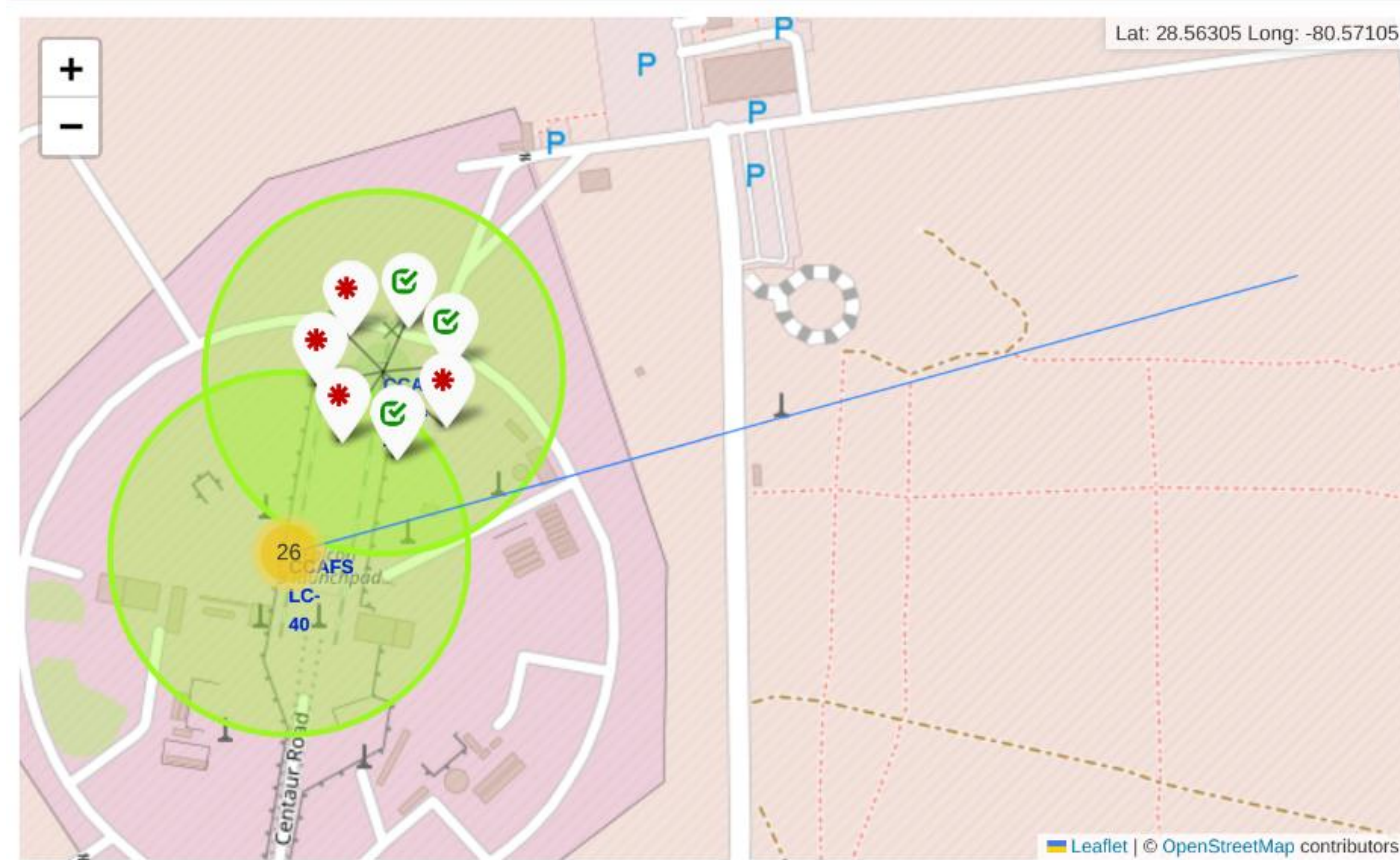
Launch Sites on map

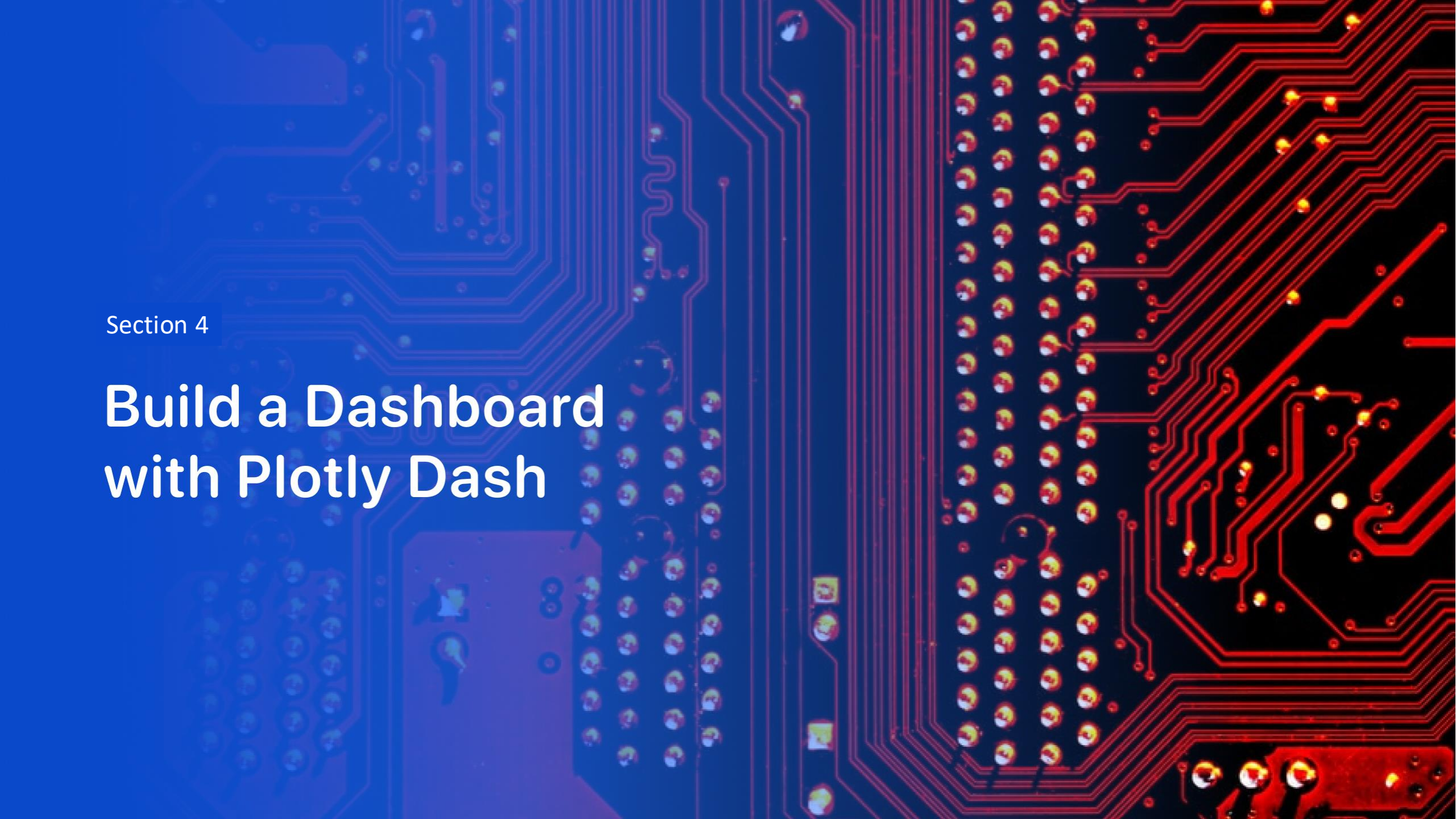
- Added a circles and markers for each unique launch site locations
- The map shows Nasa in the middle of the four launch sites, close to the shore of Texas. VAFB is on the west coast, while CCAFS LC-40, CCAFS SLC-40 and KSC LC-39A are on the east side of Florida.
- We can see that the sites are locations in the USA closest to the equator line and seas.



Launches and distances

- Folium map showing a selected launch site to its proximities with distance calculated and displayed.
- The launch sites are far from cities, roads, and building, but close to the shore as safety to people living or passing by.





Section 4

Build a Dashboard with Plotly Dash

Launch Success counts for all sites

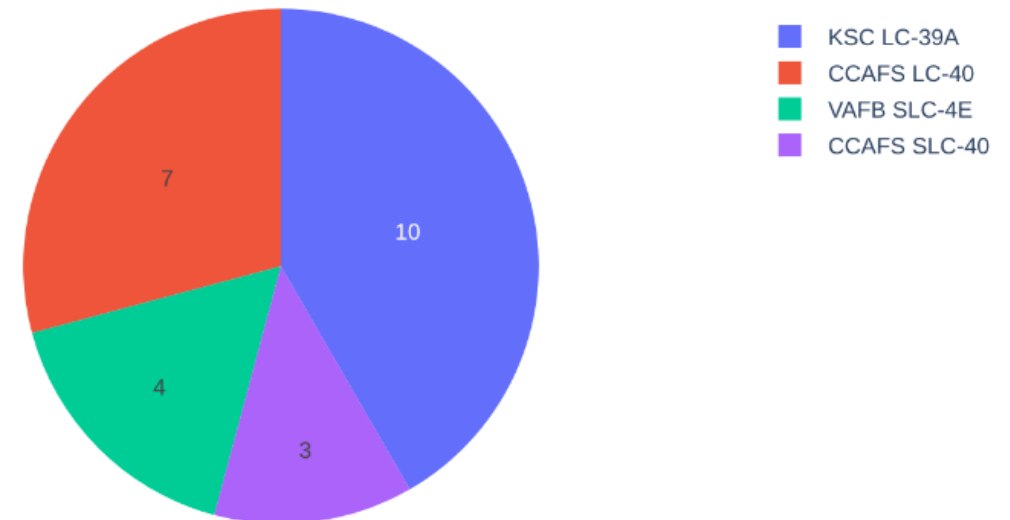
- Piechart in Dashboard showing the success counts for each launch site.
- KSC has the highest counts of success, while CCAFS SLC-40 has the lowest. Since they are close by, the location might not be an indication of success.

SpaceX Launch Records Dashboard

All Sites



Counts of launch success per site



KSC Launch site success counts

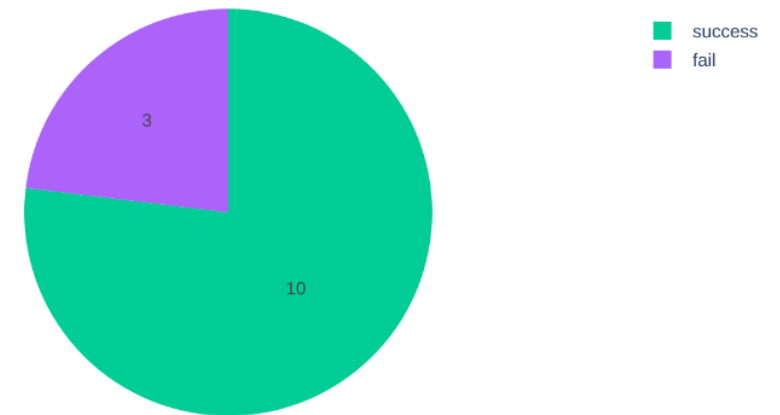
- Piechart for the launch site with highest launch success ratio: KSC LC-39A.
- The launch site also has a lower ratio of failed landing of 3 for 10 successes. We could expect that launches from this site would result in successful booster landing attempts.

SpaceX Launch Records Dashboard

KSC LC-39A



Counts of launch success in KSC LC-39A



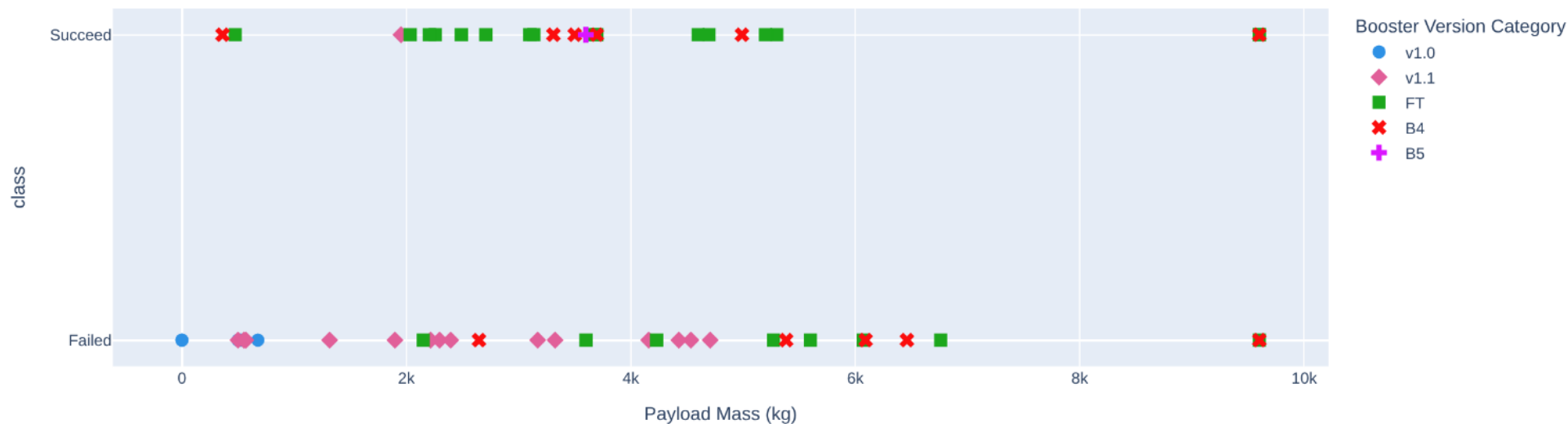
Payload Mass vs. Landing Success

- The failed landing attempts are mostly distributed for payload masses below 7k kg.
- The Booster version FT has the highest success rate for a mass range between 2k and 6k kg.
- We can see that separating in categories we get more distinctions between failure and success, so we can expect a classification algorithm to lead to good predictions.

Payload range (Kg):



Payload Mass vs. Landing Success considering booster version category



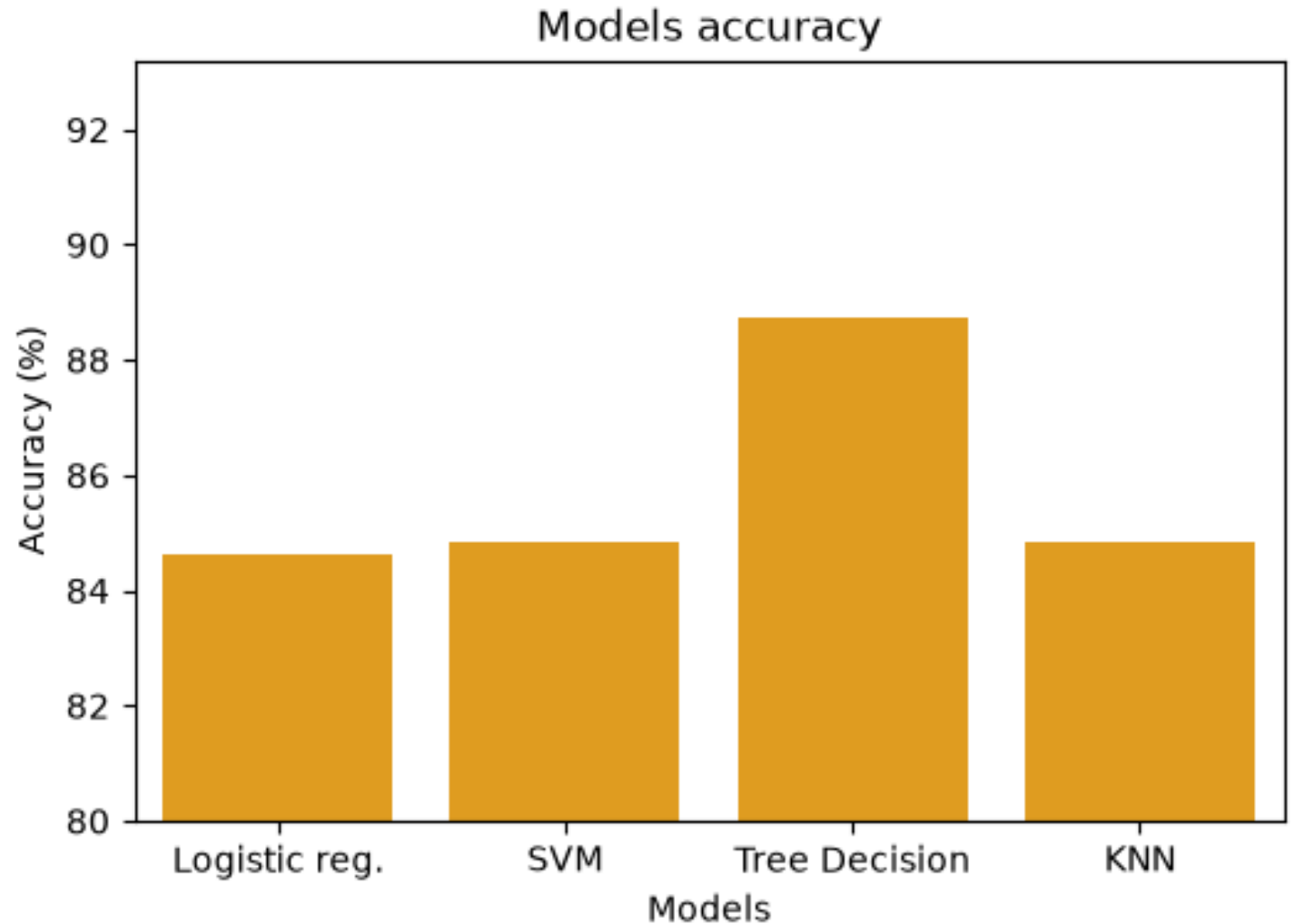


Section 5

Predictive Analysis (Classification)

Classification Accuracy

- Model with the highest classification accuracy: Decision Tree Classifier algorithm.
- Even though all the others are similar in accuracy, practically the same value of 84%, the Decision Tree can reach hyperparameters that lead to very good scores.



Confusion Matrix

- Best performing model: Decision Tree Classifier
- Accuracy of 88.75% and score of 88.89% on test data.
- As the other models it is good at predicting True-positives
- Was better at predicting True-negatives, while other models predicted same number of True and False negatives.
- Since the model is useful to predict if a SpaceX booster will not land successfully we are interested in increasing the True-negatives over False-negatives.



Conclusions

Collected data from SpaceX and Wikipedia through REST APIs

Formatted and fixed data into useful datasets

EDA revealed relationships of many features to the landing success

Checked Launch sites locations and their implications

Used classification models to get prediction of a landing success through launch data

Best model is a Decision Tree classifier with 88% accuracy.

Thank you!

